

A Practical Guide to Softmax function

TOPIC -- SOFTMAX FUNCTION

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

-- 01 --

$$P(y = j | \mathbf{x}) = \frac{e^{\mathbf{x} \cdot \mathbf{w}_j}}{\sum_{k=1}^K e^{\mathbf{x} \cdot \mathbf{w}_k}} \quad \{\displaystyle P(y=j | \mathbf{x}) = \frac{e^{\mathbf{x} \cdot \mathbf{w}_j}}{\sum_{k=1}^K e^{\mathbf{x} \cdot \mathbf{w}_k}}\}$$

-- 02 --

In words, the softmax applies the standard exponential function to each element z_i of the input tuple $\mathbf{z}$ (consisting of K real numbers), and normalizes these values by dividing by the sum of all these exponentials. The normalization ensures that the sum of the components of the output vector $\sigma(\mathbf{z})$ is 1. The term "softmax" derives from the amplifying effects of the exponential on any maxima in the input tuple. For example, the standard softmax of $(1, 2, 8)$ is approximately $(0.001, 0.002, 0.997)$, which amounts to assigning almost all of the total unit weight in the result to the position of the tuple's maximal element (of 8). In general, instead of e a different base $b > 0$ can be used. As above, if $b > 1$ then larger input components will result in larger output probabilities, and increasing the value of b will create probability distributions that are more concentrated around the positions of the largest input values. Conversely, if $0 < b < 1$ then smaller input components will result in larger output probabilities, and decreasing the value of b will create probability distributions that are more concentrated around the positions of the smallest input values. Writing $b = e^\beta$ or $b = e^{-\beta}$ (for real β) yields the expressions:

-- 03 --

$$\frac{\partial}{\partial q_k} \sigma(q, i) = \sigma(q, i) (\delta_{ik} - \sigma(q, k)) \quad \{\displaystyle \frac{\partial}{\partial q_k} \sigma(q, i) = \sigma(q, i) (\delta_{ik} - \sigma(q, k))\}$$

-- 04 --

$$P_t(a) = \frac{\exp(q_t(a)/\tau)}{\sum_{i=1}^n \exp(q_t(i)/\tau)}, \quad \{\displaystyle P_{\{t\}}(a)=\frac{\exp(q_{\{t\}}(a)/\tau)}{\sum_{i=1}^n \exp(q_{\{t\}}(i)/\tau)}\}$$

-- 05 --

We are concerned with feed-forward non-linear networks (multi-layer perceptrons, or MLPs) with multiple outputs. We wish to treat the outputs of the network as probabilities of alternatives (e.g. pattern classes), conditioned on the inputs. We look for appropriate output non-linearities and for appropriate criteria for adaptation of the parameters of the network (e.g. weights). We explain two modifications: probability scoring, which is an alternative to squared error minimisation, and a normalised exponential (softmax) multi-input generalisation of the logistic non-linearity.

-- 06 --

$$\frac{\partial}{\partial q_k} \sigma(q, i) = \sigma(q, k) (\delta_{ik} - \sigma(q, i)). \quad \{\displaystyle \frac{\partial}{\partial q_{\{k\}}} \sigma(\textbf{q}, i) = \sigma(\textbf{q}, k) (\delta_{\{ik\}} - \sigma(\textbf{q}, i)).\}$$

-- 07 --

Geometrically, softmax is constant along diagonals: this is the dimension that is eliminated, and corresponds to the softmax output being independent of a translation in the input scores (a choice of 0 score). One can normalize input scores by assuming that the sum is zero (subtract the average: $c = \frac{1}{n} \sum z_i$), and then the softmax takes the hyperplane of points that sum to zero, $\sum z_i = 0$, to the open simplex of positive values that sum to 1 $\sum \sigma(z)_i = 1$, analogously to how the exponent takes 0 to 1, $e^0 = 1$ and is positive. By contrast, softmax is not invariant under scaling. For instance, $\sigma(0, 1) = (1/(1+e), e/(1+e))$ but $\sigma(0, 2) = (1/(1+e^2), e^2/(1+e^2))$.

-- 08 --

$$\sigma_{1/100}(0, 10) = (1/(1+e^{1/10}), e^{1/10}/(1+e^{1/10})) \approx (0.475, 0.525). \quad \{\displaystyle \sigma_{\{1/100\}}(0, \{10\}) = (1/\left(1+e^{\{1/10\}}\right), e^{\{1/10\}}/\left(1+e^{\{1/10\}}\right)) \approx (0.475, 0.525).\}$$

-- 09 --

Applications The softmax function is used in various multiclass classification methods, such as multinomial logistic regression (also known as softmax regression), multiclass linear discriminant analysis, naive Bayes classifiers, and artificial neural networks. Specifically, in multinomial logistic regression and linear discriminant analysis, the input to the function is the result of K distinct linear functions, and the predicted probability for the j th class given a sample tuple x and a weighting vector w is:

-- 10 --

Statistical mechanics In statistical mechanics, the softmax function is known as the Boltzmann distribution (or Gibbs distribution): the index set $1, \dots, k$ are the microstates of the system; the inputs z_i are the energies of that state; the denominator is known as the partition function, often denoted by Z ; and the factor β is called the coldness (or thermodynamic beta, or inverse temperature).

-- 11 --

Mathematical properties Geometrically the softmax function maps the Euclidean space $\mathbb{R}^K$ to the boundary of the standard $(K - 1)$ -simplex, cutting the dimension by one (the range is a $(K - 1)$ -dimensional simplex in K -dimensional space), due to the linear constraint that all output sum to 1 meaning it lies on a hyperplane. Along the main diagonal $(x, x, \dots, x)$, softmax is just the uniform distribution on outputs, $(1/n, \dots, 1/n)$: equal scores yield equal probabilities. More generally, softmax is invariant under translation by the same value in each coordinate: adding $c = (c, \dots, c)$ to the inputs z yields $\sigma(z + c) = \sigma(z)$, because it multiplies each exponent by the same factor, e^c (because $e^{z_i + c} = e^{z_i} \cdot e^c$), so the ratios do not change: